

COM00146M – Research Methods

**Research Design for Evaluating Machine-Learning
Delivery-Delay Prediction**

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1. Research Aim and Research Questions

Delivery delays create operational costs, disrupt distribution planning and can reduce customer satisfaction. The organisation currently uses a rule-based delay-prediction system and must determine whether investment in machine learning (ML) would provide a sufficiently valuable alternative. This decision cannot depend on predictive performance alone because an ML system must also remain dependable under changing operational conditions and be usable within existing logistics workflows.

For this research, delivery-delay prediction is assumed to be a binary classification problem in which an individual shipment is classified, at a defined pre-delivery decision point, as likely to arrive on time or after its promised delivery time. This assumption is important because it determines the outcome being predicted and the evaluation metrics subsequently used.

1.1 Research Aim

The research aims to evaluate whether a machine-learning-based delivery-delay prediction system would provide a more effective, reliable and operationally suitable alternative to the organisation's existing rule-based system.

The aim is evaluative rather than confirmatory and therefore does not assume that ML will be superior. Effectiveness concerns predictive and computational performance; reliability concerns whether performance remains robust under realistic operational variation; and operational suitability concerns whether predictions can be understood, governed and incorporated responsibly into logistics decision-making.

1.2 Research Questions

RQ1: How does the predictive and computational performance of selected machine-learning models compare with the organisation's existing rule-based delivery-delay prediction system?

Previous studies demonstrate that ML can support delivery- and shipment-time prediction using operational data, while also showing that results depend on the available features, data representation and application context [1], [2]. Direct comparison against the organisation's existing rule-based system is therefore necessary rather than assuming that published ML performance will transfer to this organisation. RQ1 evaluates technical effectiveness using predictive classification measures and computational performance.

RQ2: To what extent does the performance of selected machine-learning models remain robust across time periods, operational subgroups and plausible variations in data quality?

Strong aggregate performance does not necessarily imply dependable operational performance. Predictive relationships can change over time through concept drift [3], while overall metrics may conceal poorer performance for particular routes, warehouses, transport modes or shipment types. RQ2 therefore examines whether any technical advantage found in RQ1 remains sufficiently stable under realistic operational variation.

RQ3: What operational adoption requirements do logistics stakeholders identify for the trustworthy and responsible use of machine-learning delay predictions within existing planning workflows?

Technical capability is a necessary but insufficient condition for operational adoption. Research on explainable AI in supply-chain decision-making demonstrates the relevance of explainability to operational decision support [4], while longitudinal research with logistics professionals indicates that strong AI performance alone does not guarantee appropriate trust or effective human-AI collaboration [5]. RQ3 therefore examines operational adoption through the related dimensions of explanation, human oversight, accountability, alert handling and workflow integration.

Together, the three questions address technical effectiveness, operational reliability and organisational suitability, thereby providing complementary evidence for the investment decision.

2. Research Design and Methodological Framework

2.1 Research Framework for RQ1

RQ1 requires measurable evidence of whether selected ML approaches provide a meaningful improvement over the current rule-based system. A comparative offline computational experiment within a predominantly post-positivist, deductive quantitative framework is appropriate because the research design should follow the question and required evidence rather than a preferred technique [6].

Research philosophy, approach and design

Within the module's treatment of positivist research, systematic empirical measurement is central. A post-positivist interpretation is adopted here because comparative performance can be examined through observable measures while recognising that such measurements are imperfect and conditional. Historical logistics data may contain missing or inaccurate records, predictions are probabilistic, and conclusions depend upon operational definitions, feature availability, thresholds and evaluation periods. The research therefore seeks the most defensible empirical explanation rather than treating a single measured result as certain truth [6].

The reasoning is primarily deductive. Existing literature suggests that ML can identify complex relationships within logistics data [1], [2], and the study tests whether this proposition translates into better performance in the commissioning organisation.

The independent variable is prediction approach, comprising the existing rule-based system and a small number of justified ML models. Dependent variables are predictive performance and computational performance, including precision, recall, F1-score, false-positive and false-negative rates, inference latency and throughput. Control conditions include the held-out shipments, available features, preprocessing rules, threshold-selection procedure, hardware environment and evaluation periods.

The existing rule-based system forms the operational baseline. Candidate ML approaches could include logistic regression as a comparatively interpretable benchmark and tree-based methods such as random forests and gradient boosting, which can model nonlinear relationships. Restricting evaluation to a small, justified set reduces the risk of conducting an unfocused algorithm search and subsequently reporting only the most favourable result.

Data and evaluation procedure

The organisation-provided shipment dataset constitutes secondary data because it was originally collected for operational rather than research purposes.

The definition of a delay would be fixed before modelling, for example arrival after the promised delivery time. The intended prediction point would likewise be defined in advance. Predictors might include route, transport mode, planned journey duration, warehouse-processing information, traffic and weather data, but only information available at that

decision point would be included. Information unavailable at the intended prediction point would be excluded because using outcome-related or future information during model development can create target or temporal leakage and artificially optimistic evaluation [7].

The dataset would first be profiled for missing values, duplicates, inconsistent timestamps and implausible records. Inclusion, exclusion and preprocessing rules would be documented before final evaluation so that analytical choices are not repeatedly altered according to which choices produce the strongest model performance.

Training, validation and testing would use chronological separation. Earlier records would be used for model development, a subsequent period for tuning and threshold selection, and a later unseen period for final comparison. This design more closely reflects the deployment problem of learning from past shipments to predict later ones and avoids training on information temporally downstream of the evaluation period. Temporal leakage can produce misleadingly optimistic estimates when future information influences predictions about earlier data [7]. An alternative would be unrestricted random cross-validation, which uses data efficiently and may reduce variance in estimates. However, where operational patterns are temporally dependent, random mixing of periods can weaken the realism of future-performance claims. Chronological testing is therefore preferable for the principal evaluation, although repeated temporal splits could be used to assess sensitivity.

Data analysis

Overall accuracy would not be the principal measure where delayed shipments are relatively uncommon, because aggregate accuracy can obscure minority-class performance. Precision-recall analysis is particularly informative for imbalanced binary classification [8]. Precision, recall, F1-score and area under the precision-recall curve (PR-AUC) would therefore form core measures, supported by false-positive and false-negative rates. Precision measures how many delay alerts correspond to actual delays, whereas recall measures how many genuine delays are identified. These errors have different operational consequences: false positives may increase planner workload, whereas false negatives may remove opportunities for preventive intervention.

Where probabilistic risk scores are produced, precision-recall behaviour should also be examined at alternative thresholds. Calibration would additionally assess whether predicted probabilities correspond with observed delay frequencies. Calibration would apply only to

systems producing interpretable probability estimates and therefore may not be directly comparable with a rule-based system producing only binary alerts.

Inference latency and throughput would be measured under a consistent hardware environment. Because competing systems would make predictions for the same held-out shipments, paired comparison is appropriate. Bootstrap confidence intervals could quantify uncertainty around differences between systems [9]. Statistical significance would be interpreted alongside effect size and operational relevance because a large dataset may make a very small improvement statistically detectable without making it commercially worthwhile.

Quality, alternatives and limitations

Internal validity is strengthened by evaluating competing systems on common test data under consistent preprocessing, features, hardware and evaluation procedures. Reproducibility would be supported through documented preprocessing, version-controlled code, recorded model configurations and fixed random seeds where relevant.

A credible alternative to this offline design would be a live or shadow-mode operational pilot. A live study would offer greater ecological validity because it could measure model behaviour under real data arrival, genuine infrastructure constraints and actual planner interaction. However, it would provide less experimental control, incur greater operational cost and potentially expose the organisation to an insufficiently validated system. A shadow-mode pilot, in which ML predictions are generated without influencing live decisions, could reduce this risk but would still be more complex than offline testing. Consequently, an offline experiment is preferable as the initial RQ1 design; a shadow-mode pilot would be a logical later stage if sufficient technical value is demonstrated.

The experiment remains constrained by historical data. Delay labels may be inaccurate or incomplete, and findings from one organisation and time period restrict external generalisability. Concept drift also means that future relationships may differ from historical ones [3]. Temporal testing can reveal some deterioration but cannot guarantee future stability.

Most importantly, this design establishes offline technical capability rather than organisational effectiveness. It cannot show that planners will understand or act upon predictions, nor that interventions triggered by alerts will ultimately reduce the number of late deliveries. Its conclusion must therefore remain limited: RQ1 can establish whether ML demonstrates

sufficient comparative technical performance to justify further investigation, not whether it should automatically replace the existing system.

2.2 Overall Framework for RQ1-RQ3

The wider project would adopt a sequential mixed-methods design with explanatory and complementary purposes under an overarching pragmatic position. Quantitative evidence addresses technical performance and robustness, whereas qualitative evidence addresses the situated interpretation and operational use of predictions. Mixed methods is therefore selected because the research questions require different forms of evidence, not because methodological combination is inherently superior [6], [10].

Pragmatism is appropriate because the purpose is a practical organisational decision and because it provides a methodological basis for combining qualitative and quantitative inquiry while focusing attention on the research problem and consequences of inquiry [10]. RQ1 and RQ2 are predominantly post-positivist and deductive because they involve measurable performance outcomes. RQ3 is predominantly interpretivist because it investigates how stakeholders understand trust, accountability and workflow integration.

A philosophical purist could argue that combining post-positivist and interpretivist assumptions creates inconsistency. However, the strands are not treated as interchangeable. Numerical measures establish comparative system behaviour under specified conditions, while stakeholder accounts provide contextual interpretations that those measurements cannot produce. The pragmatic framework therefore connects rather than collapses the two forms of evidence [10].

Quantitative framework for RQ1 and RQ2

RQ1 would use the comparative experiment described above.

RQ2 would extend this evaluation to robustness. First, temporal evaluation would examine performance across successive held-out periods because concept drift can undermine models trained on historical relationships [3]. Second, pre-specified subgroup analysis would compare operationally meaningful categories such as routes, warehouses, shipment types or transport modes. Minimum sample thresholds would be established so that apparent subgroup differences are not interpreted from extremely small numbers of cases.

Third, plausible data-quality stress tests could simulate realistic input degradation, such as selected missing variables, unavailable telemetry or stale operational information. Stress conditions should be specified according to known organisational failure modes rather than arbitrarily selected after model development.

Descriptive statistics, confidence intervals and paired comparisons would quantify performance and uncertainty. As with RQ1, statistical significance would be distinguished from practical significance. RQ2 can determine whether performance appears stable under the tested conditions, but it cannot guarantee future robustness under every possible operational disruption.

Qualitative case study for RQ3

RQ3 would use a qualitative case-study strategy bounded around the organisation's logistics planning and delay-management function. Case study is appropriate for investigating a contemporary organisational process within its real context [11].

Participants would be selected purposively from roles directly involved in interpreting, managing or acting upon delay information, including planners, dispatch staff, operations managers and relevant technical personnel. Purposive sampling prioritises information-rich participants rather than statistical representation [12].

Semi-structured interviews would form the principal collection method because planned themes can be combined with flexible probing of participants' experiences and interpretations [13]. Topics would include existing alert handling, false-positive and missed-delay experiences, required explanations, escalation, responsibility for intervention, human oversight and integration with existing planning systems.

A structured survey could provide broader numerical measurement of employee attitudes, but it would require relevant concerns and response categories to be known in advance and might obscure unexpected organisational issues. Given the exploratory nature of RQ3, semi-structured interviews are preferable initially.

Where access permits, interviews would be supplemented by non-participant observation of delay monitoring and alert triage. More generally, observational research can complement interview accounts by examining practice as it occurs; ethnographic research in empirical software engineering demonstrates how observation can uncover both everyday activity and

the rationalities underlying it [14]. Relevant documents, such as escalation policies, alert definitions and operating procedures, could provide a further evidence source.

Qualitative data would be analysed through a hybrid deductive-inductive thematic analysis. Thematic analysis is an analytical method rather than a research methodology [15]. Deductive coding would maintain alignment with RQ3 concepts such as trust, explanation and accountability, while inductive coding would allow unexpected organisational concerns to emerge. Analysis would follow familiarisation, initial coding, theme development, review, definition and reporting [15].

Researcher interpretation introduces subjectivity. This cannot be eliminated, but a transparent coding trail, reflexive notes and comparison across interviews, observation and documents would make interpretations more auditable. Observation could also affect employee behaviour, particularly where participants know that their work is being studied. Repeated observation and triangulation may reduce, but cannot remove, this reactivity.

The qualitative findings would be analytically rather than statistically generalisable. This limits claims about logistics organisations generally; however, the project is principally intended to support a decision within the commissioning organisation rather than estimate population-wide attitudes.

Operational records would require secure access, data minimisation and pseudonymisation where appropriate, while interview and observational participants would require informed consent, confidentiality and assurance that research data would not be used for staff performance management.

Integration

Mixed-methods research requires explicit integration rather than the parallel execution of unrelated quantitative and qualitative studies [16]. Integration would occur by connecting the stages. Quantitative findings from RQ1 and RQ2 could shape interview prompts by highlighting model disagreements, subgroup weaknesses or significant false-positive/false-negative trade-offs.

Qualitative evidence would then help interpret their operational significance. For example, improved recall accompanied by more false-positive alerts might appear technically favourable but be unacceptable if planners cannot absorb the additional workload.

Conversely, increased alert volume could be considered worthwhile where alerts allow high-value early intervention.

The final synthesis would therefore consider predictive performance, robustness, computational feasibility, explanation, trust, accountability and workflow fit. Possible recommendations could include retaining the existing system, replacing it, deploying a hybrid approach, or proceeding to a controlled shadow-mode or operational pilot. The design therefore supports a decision without presupposing that ML adoption is the correct outcome.

3. Critical Comparison of Quantitative and Qualitative Approaches

Quantitative and qualitative approaches answer different types of questions and permit different forms of conclusion. Quantitative research emphasises standardised numerical measurement, whereas qualitative research examines meanings and interpretations in context [6]. Their suitability therefore depends on what the researcher is attempting to establish.

For RQ1 and RQ2, quantitative research is preferable because the questions concern comparative performance and robustness. Applying competing systems to common shipment data permits standardised estimates of precision, recall, F1-score, PR-AUC, error rates, latency and throughput. It can support a conclusion such as: under specified historical test conditions, one approach detects a greater proportion of delayed shipments at a measured false-positive cost. Qualitative interviews could not establish that comparative technical effect reliably.

Quantitative methods also support reproducibility and can analyse large operational datasets. However, their apparent objectivity should not be overstated. Defining a delay, selecting thresholds, deciding which errors matter and selecting features are researcher and organisational choices. Numerical evidence is therefore produced within a designed analytical framework rather than existing independently of it.

Its more substantial weakness is interpretive reach. An offline experiment cannot establish that a statistically superior model is operationally usable. It may identify a higher false-positive rate but cannot determine whether that additional workload is acceptable to planners. Similarly, strong historical performance does not guarantee future reliability where operational patterns change [3].

Qualitative research is therefore more appropriate for RQ3. Semi-structured interviews can establish how stakeholders interpret alerts, explanation, responsibility and trust, while observation can identify practices that employees may not fully articulate [13], [14]. Thematic analysis can move beyond listing individual responses by identifying and interpreting patterns across the data [15].

Its strength is contextual depth, but this creates corresponding limitations. Small purposive samples cannot establish workforce-wide prevalence, and researcher interpretation influences analysis. A qualitative finding that planners distrust a form of explanation does not demonstrate that the model is technically unreliable. Stakeholder perception and technical validity therefore should not be treated as equivalent.

A counterargument is that stakeholder research is unnecessary if the organisation primarily needs to know whether ML technically outperforms its current system. This argument has some force: adequate technical performance is a necessary condition for adoption, and if ML provides no meaningful benefit then extensive implementation research may be difficult to justify. However, technical superiority is not sufficient for investment. A highly predictive system may still fail operationally if alerts cannot be acted upon, explanations are inadequate or responsibility for automated recommendations is unclear.

The opposite argument must also be considered. Stakeholder preferences should not override evidence of technical reliability. Employees may prefer familiar rules or initially distrust unfamiliar automation despite stronger measured performance. Qualitative evidence should therefore influence implementation and interpretation rather than replace technical evaluation.

Mixed methods also increases the practical demands of design, data collection, access and integration [6], [16]. A single quantitative design would be faster, whereas a purely qualitative study could explore organisational concerns in greater depth. Nevertheless, neither alone answers the complete aim. Quantitative analysis establishes whether ML provides meaningful and robust technical performance; qualitative analysis establishes the organisational conditions under which that performance could be used responsibly. Explicit integration of these findings therefore provides a more defensible basis for the investment decision than treating either technical performance or stakeholder acceptance as sufficient in isolation.

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